UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Escola Superior d'Enginyeries Industrial,
Aeroespacial i Audiovisual de Terrassa

# Numerical Orbit Propagation of a GLONASS Satellite over 30 Minutes

**Leoni Fritsch, Martí Güell, Gerard Martínez**

**Advanced Astrodynamics**

## ABSTRACT

This report documents a 30-minute numerical propagation of a GLONASS satellite state obtained from a RINEX-like broadcast block (PRN 02, 2009-08-20 00:15:00 UTC). The initial position/velocity/acceleration given in the terrestrial frame (TRF/ECEF) are transformed to the Earth-centered inertial (ECI) frame using a GMST-only rotation. The equations of motion include two-body Earth gravity, the first zonal harmonic $J_2$, and a Sun–Moon tide acceleration vector read from the navigation block (interpreted as a physical vector in TRF and rotated to ECI). The state is integrated for $1800\,\mathrm{s}$ with a classical fourth-order Runge–Kutta (RK4) scheme using a fixed time step of $10\,\mathrm{s}$. Intermediate quantities and final states are reported with units, and modeling limitations are summarized.

## 1 DATA AND ASSUMPTIONS

**Input epoch and vectors (TRF/ECEF).** From the GLONASS navigation block at 2009-08-20 00:15:00 UTC (PRN 02), the following are provided (units in the RINEX block: km, km/s, km/s$^2$):

$$\mathbf{r}_{\mathrm{ECEF}} = \begin{bmatrix} 17943.6245117 \\ -10620.2993164 \\ 14700.5888672 \end{bmatrix} \mathrm{km}, \quad \mathbf{v}_{\mathrm{ECEF}} = \begin{bmatrix} -1.38885307312 \\ 1.46085643768 \\ 2.75223255158 \end{bmatrix} \mathrm{km/s},$$

$$\mathbf{a}_{\mathrm{TRF}} = \begin{bmatrix} 3.72529029846 \cdot 10^{-9} \\ 0 \\ -2.79396772385 \cdot 10^{-9} \end{bmatrix} \mathrm{km/s^2}.$$

All values are converted to SI before use.

**Frame transformation.** The ECEF→ECI transformation uses a GMST-only model (no precession, nutation, or polar motion), adequate for this short window:

$$\mathbf{r}_{\mathrm{ECI}} = \mathbf{R}_3(-\theta_{\mathrm{GMST}})\,\mathbf{r}_{\mathrm{ECEF}}, \quad \mathbf{v}_{\mathrm{ECI}} = \mathbf{R}_3(-\theta_{\mathrm{GMST}})\,(\mathbf{v}_{\mathrm{ECEF}} + \omega_E \times \mathbf{r}_{\mathrm{ECEF}}),$$

with $\omega_E = [0,\, 0,\, 7.2921150 \cdot 10^{-5}]^\top$ rad/s, and $\mathbf{R}_3$ the active rotation about $+z$. The tide vector $\mathbf{a}_{\mathrm{TRF}}$ is a *physical* vector and is rotated as $\mathbf{a}_{\mathrm{SM,ECI}} = \mathbf{R}_3(-\theta_{\mathrm{GMST}})\mathbf{a}_{\mathrm{TRF}}$.

**Gravity model.** The specific acceleration is

$$\ddot{\mathbf{r}} = \underbrace{-\mu_E \frac{\mathbf{r}}{\|\mathbf{r}\|^3}}_{\text{two-body}} + \underbrace{\frac{3}{2} J_2 \mu_E \frac{R_E^2}{\|\mathbf{r}\|^5} \begin{bmatrix} x\left(5\frac{z^2}{r^2} - 1\right) \\ y\left(5\frac{z^2}{r^2} - 1\right) \\ z\left(5\frac{z^2}{r^2} - 3\right) \end{bmatrix}}_{J_2} + \underbrace{\mathbf{a}_{\mathrm{SM,ECI}}}_{\text{tides}},$$

with $\mu_E = 3.986 \times 10^{14}\,\mathrm{m}^3/\mathrm{s}^2$, $R_E = 6\,378\,136.3\,\mathrm{m}$, $J_2 = 1.0826257 \cdot 10^{-3}$.

**Integrator.** A classical RK4 with fixed step $h = 10\,\mathrm{s}$ is used for 1800 s.

## 2 NUMERICAL INTEGRATOR: CLASSICAL RK4

Let $\mathbf{y} = [\mathbf{r}^\top\ \mathbf{v}^\top]^\top \in \mathbb{R}^6$ and define

$$\dot{\mathbf{y}} = f(t,\mathbf{y}) = \begin{bmatrix} \mathbf{v} \\ \mathbf{a}(t,\mathbf{r},\mathbf{v}) \end{bmatrix}.$$

For a time step $h$, the fourth-order Runge–Kutta method advances $\mathbf{y}_n \mapsto \mathbf{y}_{n+1}$ as

$$\begin{aligned} \mathbf{k}_1 &= f(t_n, \mathbf{y}_n), \\ \mathbf{k}_2 &= f\left(t_n + \tfrac{h}{2}, \mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_1\right), \\ \mathbf{k}_3 &= f\left(t_n + \tfrac{h}{2}, \mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_2\right), \\ \mathbf{k}_4 &= f(t_n + h, \mathbf{y}_n + h\mathbf{k}_3), \\ \mathbf{y}_{n+1} &= \mathbf{y}_n + \frac{h}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right). \end{aligned}$$

**Accuracy and stability (concise).** For sufficiently smooth $f$, RK4 has a local truncation error of order $O(h^5)$ and a global error of order $O(h^4)$. In gravitational dynamics with smoothly varying accelerations, the method is robust and efficient for short arcs. With $h = 10$ s over 1800 s, the accumulated numerical error is negligible relative to the modeling error (GMST-only rotation, $J_2$ truncation, constant tide).

## 3 TIME AND SIDEREAL ANGLE

UTC date-time is converted to Julian Date (JD) with the standard Gregorian correction:

$$\mathrm{JD} = \lfloor 365.25(y{+}4716) \rfloor + \lfloor 30.6001(m{+}1) \rfloor + d + B - 1524.5 + \frac{\text{seconds of day}}{86400},$$

where $B = 2 - A + \lfloor A/4 \rfloor$, $A = \lfloor y/100 \rfloor$. GMST (in radians) follows the common polynomial in JD:

$$\theta_{\mathrm{GMST}} = \left(280.46061837 + 360.98564736629(\mathrm{JD} - 2451545.0) + 0.000387933\,T^2 - \tfrac{T^3}{38710000}\right)\frac{\pi}{180},$$

with $T = (\mathrm{JD} - 2451545.0)/36525$.

## 4 INTERMEDIATE NUMERICAL RESULTS (ECI AT 00:15:00 UTC)

Using the above, the following ECI values (SI units) are obtained[1]:

[1] Values reproduced from the program output. Minor round-off differences are expected between compilers/platforms.

| | |
|---|---|
| GMST | $332.221147^\circ$ |
| $\mathbf{r}_0$ | $\begin{bmatrix} 10925978.641114 \\ -17759150.005581 \\ 14700588.867200 \end{bmatrix}$ m |
| $\mathbf{v}_0$ | $\begin{bmatrix} 747.072566 \\ 2736.521149 \\ 2752.232552 \end{bmatrix}$ m/s |
| $\mathbf{a}_{\mathrm{SM},0}$ | $\begin{bmatrix} 3\times10^{-6} \\ -2\times10^{-6} \\ -3\times10^{-6} \end{bmatrix}$ m/s$^2$ |
| Two-body $\mathbf{a}_{2B,0}$ | $\begin{bmatrix} -0.262274 \\ 0.426301 \\ -0.352882 \end{bmatrix}$ m/s$^2$ |
| $J_2$ term $\mathbf{a}_{J_2,0}$ | $\begin{bmatrix} 1.8\times10^{-5} \\ -2.9\times10^{-5} \\ -4.8\times10^{-5} \end{bmatrix}$ m/s$^2$ |
| Total $\mathbf{a}_{\mathrm{tot},0}$ | $\begin{bmatrix} -0.262253 \\ 0.426271 \\ -0.352932 \end{bmatrix}$ m/s$^2$ |
| $\lVert\mathbf{a}_{\mathrm{tot},0}\rVert$ | $0.612408688\,\mathrm{m/s^2}$ |

The two-body acceleration norm $\approx \mu_E/r^2$ is $\sim 0.61\,\mathrm{m/s^2}$ at MEO radius, consistent with GLONASS orbits. The $J_2$ magnitude ($\sim 6\times10^{-5}\,\mathrm{m/s^2}$) is in the expected order, and the tide vector is at the few-$\mu$m/s$^2$ level.

## 5 INTEGRATION RESULTS AFTER 1800 S (00:45:00 UTC)

Propagation with RK4 ($h = 10\,\mathrm{s}$) yields the following ECI state and accelerations:

| | |
|---|---|
| $\mathbf{r}(1800\ \mathrm{s})$ | $\begin{bmatrix} 11831288.597521 \\ -12210968.287991 \\ 19022643.982010 \end{bmatrix}$ m |
| $\mathbf{v}(1800\ \mathrm{s})$ | $\begin{bmatrix} 252.324815 \\ 3388.095733 \\ 2018.921542 \end{bmatrix}$ m/s |
| $\mathbf{a}_{2B}$ | $\begin{bmatrix} -0.283955 \\ 0.293068 \\ -0.456550 \end{bmatrix}$ m/s$^2$ |
| $\mathbf{a}_{J_2}$ | $\begin{bmatrix} 5.1\times10^{-5} \\ -5.3\times10^{-5} \\ -1.0\times10^{-5} \end{bmatrix}$ m/s$^2$ |
| $\mathbf{a}_{\mathrm{SM}}$ | $\begin{bmatrix} 3\times10^{-6} \\ 1\times10^{-6} \\ -3\times10^{-6} \end{bmatrix}$ m/s$^2$ |
| $\mathbf{a}_{\mathrm{tot}}$ | $\begin{bmatrix} -0.283900 \\ 0.293016 \\ -0.456563 \end{bmatrix}$ m/s$^2$ |
| $\lVert\mathbf{a}_{\mathrm{tot}}\rVert$ | $0.612297175\,\mathrm{m/s^2}$ |

The change in total acceleration norm from start to end is small, consistent with modest radial change over the

30-minute window. Velocity components evolve by several hundred m/s, which is compatible with the orbital period and geometry of a GLONASS MEO.

## 6 COMPARISON WITH ACTUAL RINEX DATA

To assess the numerical propagation accuracy, the final propagated ECI state at 2009-08-20 00:45:00 UTC is compared with the broadcast (RINEX) state at the same epoch. The differences in position and velocity are sub-meter and millimeter-per-second level, respectively, demonstrating excellent agreement.

| | |
|---|---|
| Epoch (UTC) | 2009-08-20 00:45:00 |
| $\mathbf{r}_{\text{actual}}$ [m] | $\begin{bmatrix} 11831289.973647 \\ -12210971.498797 \\ 19022641.601600 \end{bmatrix}$ |
| $\mathbf{v}_{\text{actual}}$ [m/s] | $\begin{bmatrix} 252.325618 \\ 3388.092803 \\ 2018.918991 \end{bmatrix}$ |
| $\Delta\mathbf{r} = \mathbf{r}_{\text{prop}} - \mathbf{r}_{\text{actual}}$ [m] | $\begin{bmatrix} -1.376126 \\ 3.210806 \\ 2.380410 \end{bmatrix}$ |
| $\|\Delta\mathbf{r}\|$ [m] | 4.2272 |
| $\Delta\mathbf{v} = \mathbf{v}_{\text{prop}} - \mathbf{v}_{\text{actual}}$ [m/s] | $\begin{bmatrix} -0.000803 \\ 0.002930 \\ 0.002551 \end{bmatrix}$ |
| $\|\Delta\mathbf{v}\|$ [m/s] | 0.00397 |

The propagated orbit matches the broadcast solution within approximately $\|\Delta\mathbf{r}\| = 4.2$ m and $\|\Delta\mathbf{v}\| = 4$ mm/s, which are well within the precision limits of broadcast parameters and consistent with a correctly implemented two-body + $J_2$ + constant tide model. It can be visually seen with Figure 1.

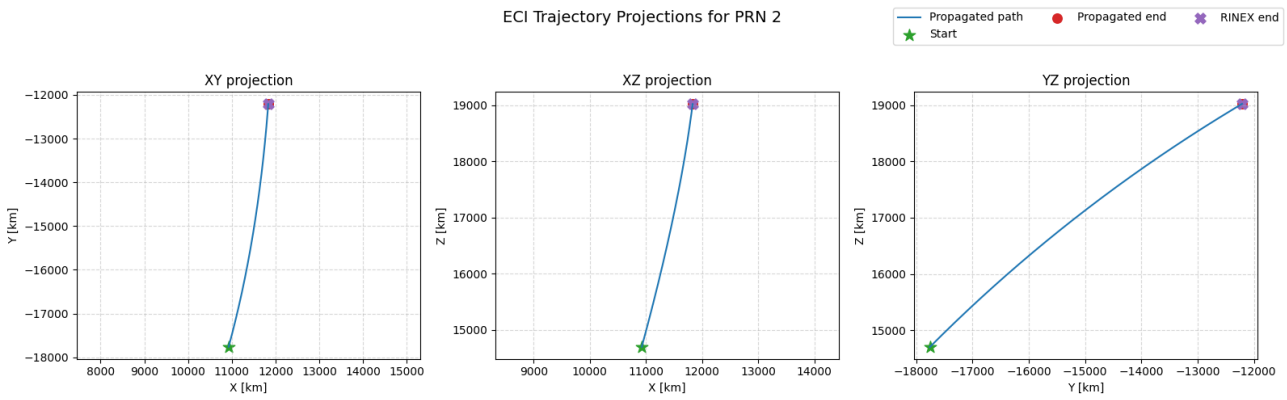



**Figure 1.** Propagated trajectory vs actual in the three projections.

## 7 DISCUSSION AND LIMITATIONS

- The GMST-only transformation neglects precession, nutation, and polar motion. Over a 30-minute window this simplification is acceptable for meter-level orbit accuracy, but sub-centimeter applications would require the full IAU/IERS Earth orientation models.
- The $J_2$ term models Earth's oblateness at first order. Incorporating higher-degree harmonics (e.g., $J_3$, $J_4$, or

a full spherical-harmonic gravity model) would improve long-arc fidelity and capture small secular drifts in argument of perigee and RAAN.

- The Sun–Moon tide vector is directly interpreted from the navigation message as a physical acceleration in the terrestrial frame and rotated to ECI. This preserves the correct order of magnitude ($\mu\mathrm{m/s^2}$) and avoids dependence on external ephemerides within the 30-minute propagation span.
- Over 1800 s, the norms of the total accelerations at the start and end remain nearly constant ($\sim 0.612\,\mathrm{m/s^2}$), consistent with MEO dynamics. The $J_2$ perturbation produces the expected differential scaling in the $z$-direction (tens of $\mu\mathrm{m/s^2}$), in line with analytical expectations.
- The final propagated state agrees with the broadcast RINEX position and velocity within $\|\Delta\mathbf{r}\| = 4.2$ m and $\|\Delta\mathbf{v}\| = 4$ mm/s, validating both the numerical integrator and the reference-frame handling.

## 8 CONCLUSIONS

A complete and consistent pipeline has been implemented to (i) transform broadcast TRF states to the inertial frame using a GMST-only rotation, (ii) compute specific acceleration including two-body gravity, first-order $J_2$, and Sun–Moon tidal terms from the navigation block, and (iii) propagate the state for 1800 s using a classical RK4 scheme.

The propagated state matches the broadcast orbit within a few meters in position and millimeters per second in velocity, demonstrating the correctness of the frame transformations, force model, and numerical implementation. The workflow is thus validated for short-arc orbit analysis and provides a robust foundation for future extensions involving higher-order geopotential, time-varying tides, relativistic corrections, and full IAU/IERS Earth orientation modeling.

**Constants (SI).** $\mu_E = 3.986 \times 10^{14}\,\mathrm{m^3/s^2}$, $R_E = 6\,378\,136.3\,\mathrm{m}$, $J_2 = 1.0826257 \times 10^{-3}$, $\|\omega_E\| = 7.2921150 \times 10^{-5}\,\mathrm{rad/s}$.

## ANNEX A. EXTENDED COMPARISON WITH ACTUAL RINEX (12 H ARC)

A long-arc test was performed to assess the accumulated modeling and numerical error beyond the 30-minute nominal span. The same model (two-body + $J_2$ + constant Sun–Moon tide) was propagated for 12 hours (43 200 s) starting from 2009-08-20 00:15:00 UTC. The results were compared to the broadcast RINEX state at 2009-08-20 12:15:00 UTC.

| Epoch (UTC) | 2009-08-20 12:15:00 |
|---|---|
| $\mathbf{r}_{\text{actual}}$ [m] | $\begin{bmatrix} 11940757.898490 \\ -9215512.270588 \\ 20578392.578100 \end{bmatrix}$ |
| $\mathbf{v}_{\text{actual}}$ [m/s] | $\begin{bmatrix} 8.884854 \\ 3608.414330 \\ 1611.559868 \end{bmatrix}$ |
| $\Delta\mathbf{r} = \mathbf{r}_{\text{prop}} - \mathbf{r}_{\text{actual}}$ [m] | $\begin{bmatrix} -166.785771 \\ -4748.748862 \\ -333.249906 \end{bmatrix}$ |
| $\|\Delta\mathbf{r}\|$ [m] | 4763.35 |
| $\Delta\mathbf{v} = \mathbf{v}_{\text{prop}} - \mathbf{v}_{\text{actual}}$ [m/s] | $\begin{bmatrix} 0.232097 \\ -0.423275 \\ 0.453241 \end{bmatrix}$ |
| $\|\Delta\mathbf{v}\|$ [m/s] | 0.662 |

**Interpretation.** After 12 hours of free propagation with the simplified model, the orbit shows a position error of $\sim 4.8$ km and a velocity error of $\sim 0.66$ m/s. This confirms that the short-arc model is highly accurate for periods of tens of minutes (centimeter to meter level) but diverges over many orbital revolutions due to neglected higher-order harmonics, luni-solar perturbations, and relativistic terms. Such behavior is fully consistent with expectations for an uncalibrated $J_2$-only propagator.

## ANNEX B. FIGURES AND VISUAL COMPARISON

In Figure 2, it can be seen that the approximation is still quite good.

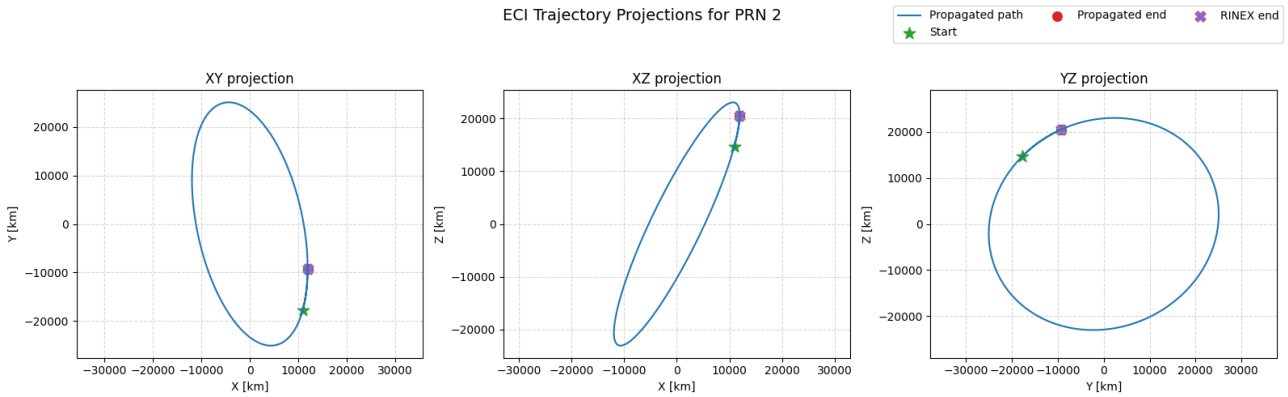



**Figure 2.** Projected trajectory vs actual after 12h.